

Key: Draw the Line Somewhere Else

84-355 Democracy's Data | Areal units | Instructor copy — do not distribute

Jonathan Cervas

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All code is given to students. Grade the interpretation.

Structure. The brief (`areal-units-brief.Rmd`) and the lab (`areal-units.Rmd`) run the same numbered steps — the brief in prose with four figures, the lab with the code exposed. **Students should read the brief first.** Step numbers below refer to both.

This is the conceptual spine of the redistricting half of the course. Every lab that measures a group by area — `redistricting`, `vote-dilution`, `rpv`, `redlining` — inherits the problem demonstrated here. Run it before them if the calendar allows.

Every number below was recomputed against the built files while this document was knitted. Nothing is typed.

The session in one sentence

The same people, counted on the same day, divided into six units of the same size, yield anywhere from 0 to 4 majority-Black units depending only on the angle of the cuts — so “how many majority-Black districts does this county support?” is not a question about the county.

The findings

```
c(fulton_people      = FN("ful_pop"),
  fulton_black_pct   = FN("ful_black_pct"),
  plans_built        = FN("n_plans"),
  people_per_unit     = FN("quota"),
  worst_deviation_pct = FN("max_dev_pct"),
  majority_black_min  = FN("maj_min"),
  majority_black_max  = FN("maj_max"),
  contiguous_plans    = FN("plans_contig"),
  contiguous_maj_min  = FN("contig_maj_min"),
  contiguous_maj_max  = FN("contig_maj_max"))
```

```

#>      fulton_people    fulton_black_pct    plans_built    people_per_unit
#>      1066710.00          44.78          360.00          177785.00
#> worst_deviation_pct majority_black_min majority_black_max contiguous_plans
#>           0.71           0.00           4.00           172.00
#> contiguous_maj_min contiguous_maj_max
#>           2.00           3.00

```

- **1,066,710 people, 11,748 blocks, one census.**
- **The scale ladder:** weighted spread 37.9 points across blocks, 36.2 across tracts, 0.0 across the county. The **weighted mean never moves:** 44.78% on every rung.
- **Statewide**, the share of Black Georgians living in a majority-Black unit falls from 66.3% (blocks) to 55.4% (tracts) to 34.8% (counties). Nobody moved.
- **The correlation with under-18 share changes sign:** 0.088 across blocks, -0.147 across counties.
- **The zoning sweep:** 360 equal-population partitions, worst deviation 0.71% of the quota, giving 0 to 4 majority-Black units. Restricted to the 172 effectively contiguous plans, still 2 to 3.

The three things to get said

1. Scale and zoning are different effects, and the second is the one that matters. Students arrive knowing “bigger units smooth things out” and will try to file the whole lab under that. Steps 3–5 are the scale effect and they are the warm-up. **Step 6 holds scale exactly fixed** — same number of units, same population in each, to 0.71% — and the answer still moves by 4 seats. Make somebody say back to you why that is not the same phenomenon.

2. The mean is the control, and nobody notices it. In Step 4 the population-weighted mean is 44.78% at every rung, exactly. **Point at that column before anything else.** It is what makes the rest an experiment rather than an anecdote: one quantity is provably unchanged, so everything that did change is attributable to the unit. Students who miss this think the lab is showing that aggregation introduces error. It isn’t. The arithmetic is exact throughout.

3. A race-blind rule has a racial output. Neither the strip rule nor the split rule reads a single race variable. One produces 0 majority-Black units, the other 4. **This is the sentence to get on the board**, because “the algorithm doesn’t look at race” is the most common defence of automated redistricting and this lab shows it establishes nothing at all about the outcome.

The line for the board: *the map is not a measurement of the people; it is half of the measurement.*

Walking the steps

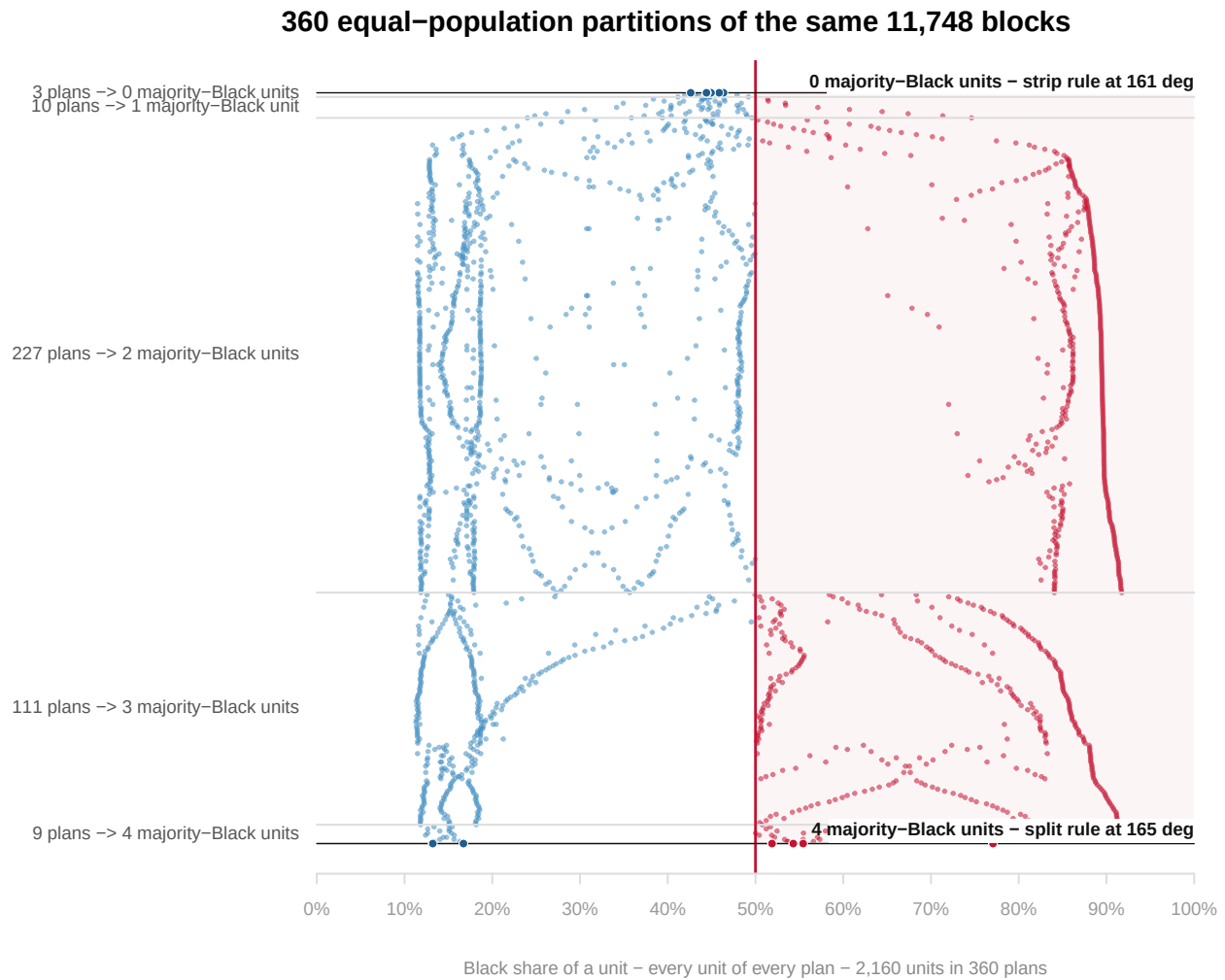
Step	What it does	Min	Watch for
1 The same people	1,066,710 people, 11,748 blocks	4	Say “any part Black” out loud and say it is a choice. Someone will ask; it is 33.03% vs 31.00% statewide.

Step	What it does	Min	Watch for
2 GEOID as a string	Block group and tract are prefixes	5	Do not skip. It is what makes Steps 3-5 airtight: no spatial operation, so no geometric error to blame.
3 Scale effect, seen	Four maps, one ramp	8	Let them look before you talk. Ask what happened to the black neighbourhoods between panel 1 and panel 4.
4 Scale effect, measured	The ladder	9	The constant mean. Also the sign flip in the correlation column.
5 Distribution collapse	Density by rung	7	It does not collapse much for Fulton. That is the finding, not a failure.
6 The zoning effect	360 partitions	14	The pivot. Take the prediction first (below).
7 The two extremes	The maps of both	9	Name <i>cracking</i> only after they have seen it. Stress that no rule saw race.
8 Are they districts?	Contiguity	8	The honest objection, honestly answered. 0 of 360 pass strictly, and the reason is rivers, not shapes.
9 Why it is a democracy problem	Section 2	8	“There is no true value to recover” is the hard idea. Give it room.
10 The ecological fallacy	The tension	5	Name rpv as the answer-key case. Do not resolve it.
11 What you can say	Including what you cannot	4	Close on the range, not on a number.

Step 6 deserves the prediction. Before showing the figure, put the setup on the board — same 11,748 blocks, 6 units, everyone within 0.71% of 177,785 people, two rules neither of which reads race — and ask: **how many of the six will be majority-Black?** Almost everyone gives one number, usually two or three, because the county is 44.78% Black. The answer is that 3 plans give 0 and 9 give 4. The room reliably assumes the question has an answer.

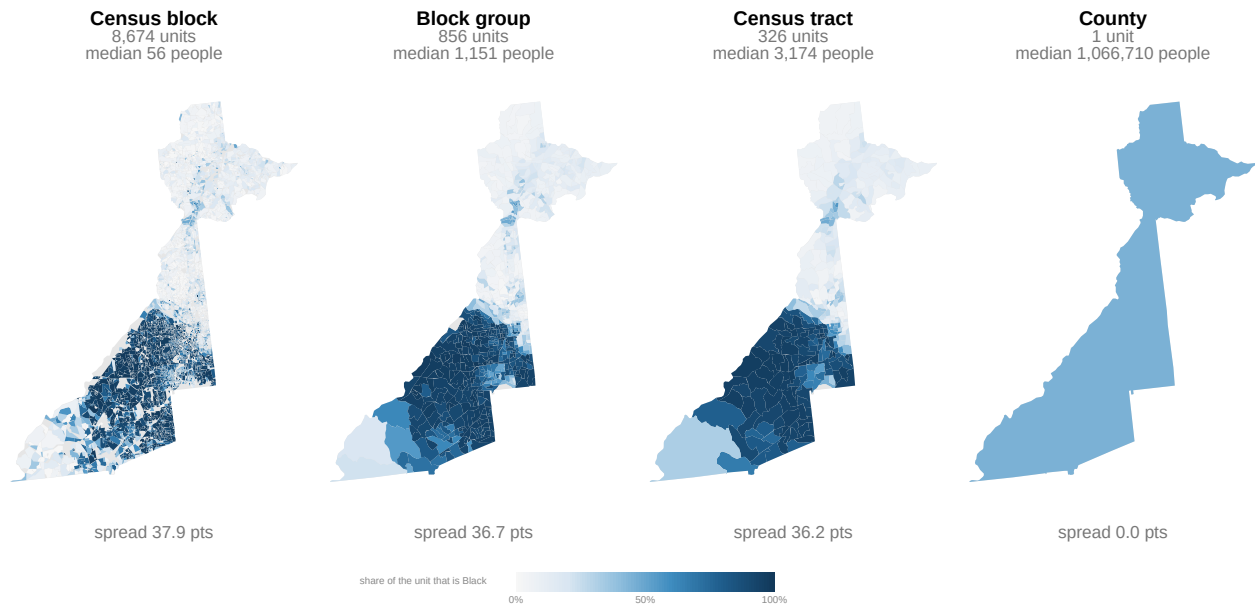
If you are short on time, protect Steps 4, 6 and 8. The constant mean, the sweep, and the contiguity test that fails to rescue us are the argument.

The figure to project



If you show one image, show this one. The red band is above 50%; each row is one plan; the two black rows are the extremes. **The whole argument is that every row contains the same 1,066,710 people.**

And the same county at four scales, for Step 3:



Handling the room

The thing that goes wrong is that a few students conclude the data is worthless — “so you can make it say anything.” That is the wrong lesson and it is worth pushing back on directly:

- **The census count is exact and the arithmetic is exact.** Nothing here is measurement error. Say so.
- **The ambiguity is bounded and measurable.** 0 to 4, and 2 to 3 among plausible plans. That is a *result*, not a shrug.
- **Some questions are unaffected.** “How many Black people live in Fulton County?” has one answer: 477,624. Ask the room which of their own questions are unit-dependent and which are not. That sorting exercise is the most useful five minutes in the session.

The second thing is students conflating this with the ecological fallacy, which they may have met in *levels-of-aggregation*. They are cousins, not twins. The ecological fallacy is about inferring *individuals* from areas. The modifiable areal unit problem is about the *area statistic itself* being non-unique before anyone infers anything. Step 10 names both; keep them apart.

Option A — A different county

Verified output

```
CTY <- c("Fulton" = "13121", "Chatham" = "13051", "Dougherty" = "13095",
        "Forsyth" = "13117", "Clayton" = "13063")
o <- do.call(rbind, lapply(names(CTY), function(nm) {
```

```

d <- bl[bl$county == CTY[[nm]], ]
z <- ladder(d)
data.frame(county = nm, people = n(sum(d$pop)), black_pct = z$mean_share[1],
           sd_blocks = z$sd_share[1], sd_tracts = z$sd_share[3],
           collapse_pts = round(z$sd_share[1] - z$sd_share[3], 1),
           in_maj_blocks = z$pct_black_in_majority[1],
           in_maj_tracts = z$pct_black_in_majority[3])
)))
o

```

County	People	Black pct	Sd blocks	Sd tracts	Collapse pts	In maj blocks	In maj tracts
Fulton	1,066,710	44.78	37.9	36.2	1.7	79.6	77.6
Chatham	295,291	39.10	31.1	24.0	7.1	66.7	55.6
Dougherty	85,790	71.64	27.7	20.5	7.2	90.2	88.0
Forsyth	251,283	5.26	5.1	2.5	2.6	0.1	0.0
Clayton	297,595	72.70	21.9	14.4	7.5	91.7	95.1

3/4: Runs a second county, reports its ladder, notices the spread falls there too.

4/4: Compares the *size* of the collapse. Fulton loses 1.7 points of spread between blocks and tracts; Chatham loses 7.1. Explains why: **the scale effect bites in proportion to how much of the segregation happens below the size of a tract.** Fulton's is a north-south divide tens of kilometres across and survives aggregation; Chatham's is finer-grained and does not.

4/4, best: Notices Clayton, where `pct_black_in_majority` goes the *wrong* way — 91.7% at block level rising to 95.1% at tract level. In a county that is already 72.7% Black, coarsening *pulls people into* majority-Black units instead of out of them. **The scale effect has no fixed direction; it depends on which side of 50% the surrounding area sits.** A student who works that out has understood the mechanism rather than memorised the Fulton result. Forsyth is the mirror image: at 5.3% Black, exactly 2 of its 1,282 populated blocks are majority-Black, and none of its tracts are.

Option B — A different definition of “Black”

Verified output

```

d <- bl[bl$county == "13121", ]
z <- rbind(cbind(measure = "any part Black", ladder(d, "black_any")),
          cbind(measure = "Black alone", ladder(d, "black_alone")))
z[, c("measure", "level", "units", "mean_share", "sd_share",
      "units_majority", "pct_black_in_majority")]

```

Measure	Level	Units	Mean share	Sd share	Units majority	Pct black in majority
any part Black	block	8674	44.78	37.9	4129	79.6
any part Black	block group	856	44.78	36.7	361	77.9
any part Black	tract	326	44.78	36.2	135	77.6

Measure	Level	Units	Mean share	Sd share	Units majority	Pct black in majority
Black alone	block	8674	42.55	37.7	3837	79.7
Black alone	block group	856	42.55	36.3	351	78.6
Black alone	tract	326	42.55	35.8	134	79.0

3/4: Reports that the numbers are lower under “Black alone” and that the pattern is the same.

4/4: Notices *which* statistic is sensitive. The mean falls 2.23 points, and the **count of majority-Black blocks falls from 4,129 to 3,837** — about 7.1% — while the spread barely moves (37.9 to 37.7 points). **Threshold counts are far more sensitive to the definition than distributional statistics are**, because a definition change moves units across a line rather than rearranging them.

4/4, best: Connects it to Step 6. If a 2.23-point change in *definition* can move the count of majority-Black units, then the 4-unit range from *zoning* is not the only non-uniqueness in the answer — and the two compound. Notes that litigants choose the definition too, and that the choice is not neutral.

Option C — A different threshold

Verified output

```
k <- paste(sw$family, sw$angle)
o <- do.call(rbind, lapply(c(40, 45, 50, 55, 60, 70), function(t)
  do.call(rbind, lapply(c("share", "bvap"), function(v) {
    cnt <- tapply(sw[[v]] > t, k, sum)
    data.frame(threshold = paste0(t, "%"),
               measure = c(share = "total population", bvap = "voting age")[[v]],
               fewest = min(cnt), most = max(cnt), range = max(cnt) - min(cnt),
               plans_at_fewest = sum(cnt == min(cnt)))
  })))
o
```

Threshold	Measure	Fewest	Most	Range	Plans at fewest
40%	total population	2	6	4	46
40%	voting age	2	6	4	70
45%	total population	1	4	3	1
45%	voting age	0	4	4	1
50%	total population	0	4	4	3
50%	voting age	0	4	4	4
55%	total population	0	3	3	7
55%	voting age	0	3	3	8
60%	total population	0	3	3	10
60%	voting age	0	3	3	11
70%	total population	0	2	2	24
70%	voting age	0	2	2	29

3/4: Reports that the counts change with the threshold, and that voting-age numbers are a little lower than total-population numbers.

4/4: Reads the **range** column, which is the point: **at no threshold does the choice of plan stop mattering**. Even at 70% the gap between the best and worst plan is 2 units. There is no threshold at which the map becomes irrelevant.

4/4, best: Notes that the ambiguity is *largest* right where the law puts its line. At 50% of total population the range is 4 of 6 units; the biggest gap anywhere in the table is 4. **The legal threshold sits in the region of maximum sensitivity**, which is not a coincidence — 50% is near the county’s own 44.78%, and a threshold near the population share is exactly where partitioning has the most leverage. A student who says that has found something the lab does not state.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is given, and this lab gives more of it than most.
- **The move that earns a 4** is treating “the Black share of a district” as a joint fact about people and boundaries, and reasoning about the *range* rather than reporting a number.
- **A 2** is “so the data can say anything” — nihilism dressed as sophistication. The ambiguity here is measured, bounded, and smaller than the student thinks.
- **Also a 2:** describing the scale effect (Steps 3-5) and stopping, having missed that Step 6 holds scale fixed.
- **Also a 2:** treating the zoning result as evidence that someone gerrymandered Fulton. Nothing in the lab has intent in it.
- **Reward anyone who asks what the range would be over *all* partitions**, not just these two rule families. The honest answer is that we do not know and it is certainly wider; noticing that 360 is a convenience sample of an enormous space is the most sophisticated available response.

Anticipated questions

- “*Isn’t this just saying bigger units are smoother?*” — No, and Step 6 is the proof: the units there are all the same size. Have this answer ready; it is the most common misreading.
- “*Which of the 360 plans is the right one?*” — None. That is the finding. A real plan is chosen by a political process operating under legal constraints, and the constraints do not narrow it to one.
- “*Couldn’t a computer just find the fairest map?*” — Only once “fairest” is defined, and the definitions conflict. Point at the two rules: both are perfectly principled and they disagree by 4 seats.
- “*Why does contiguity fail for every plan?*” — Rivers, interstates and the airport are their own blocks and touch nothing across them. The strict test is measuring the block layer, not the plan. Hence the 99%-of-population criterion, which is stated in Step 8 rather than hidden.
- “*Is 0.71% population deviation realistic?*” — It is far tighter than required. Congressional plans aim for near-zero; local plans are commonly allowed 10% total deviation, which would make the range found here **wider**, not narrower.
- “*Does this mean census tracts are wrong?*” — No. Tracts are a reasonable unit for many questions. They are the wrong unit for any question whose answer depends on where a 50% line falls.

Connections

- **precinct-geography** — the same machinery with **time** as the variable instead of the unit: boundaries move between elections. Run the two together and the pair covers both ways a unit can betray you.

- **levels-of-aggregation** — meets the modifiable areal unit problem in passing, with a random-regrouping null. This lab is that observation made the whole subject, and adds the zoning effect, which that lab does not isolate.
- **redistricting** — the applied version. Everything there assumes a plan; this lab is about the space of plans it was drawn from.
- **vote-dilution and rpv** — both measure race by area and inherit this problem wholesale. **rpv** is also the one place in the course where the ecological inference can be checked against individual records.
- **redlining** — point-in-polygon and the same assumption that people are spread evenly inside a polygon, in a setting where the error cannot be measured.
- **census-geography** — the nesting spine that Step 2 exploits, and the list of things that do not nest.